

KUTCHAN, Japan

WMO: 474330

Lat: 42.9011N

Lon: 140.7581E

Elev: 188

StdP: 99.09

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-13.0	-17.2	0.8	-14.7	-15.1	1.0	-12.4	11.7	-1.0	10.0	-2.9	1.3	50	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	28.0	22.3	26.4	21.5	25.1	20.7	23.5	26.5	22.7	25.3	21.9	24.2	4.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	17.6	25.3	21.8	16.8	24.5	21.0	16.0	23.7	71.2	26.6	68.2	25.3	65.1	24.4	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.4	7.3	-19.9	31.1	2.3	1.5	-21.5	32.2	-22.8	33.1	-24.1	33.9	-25.8	35.0		
			-20.1	24.6	2.3	1.1	-21.7	25.4	-23.1	26.1	-24.3	26.7	-26.0	27.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-5.9	-5.2	-1.4	4.8	11.4	15.9	20.2	21.1	16.5	9.6	2.8	-3.4		
	DBStd	10.04	3.24	3.39	3.43	3.46	3.17	2.87	2.50	2.58	3.27	3.33	4.00	3.51		
	HDD10.0	2137	493	426	354	160	22	1	0	0	1	47	218	416		
	HDD18.3	4226	752	659	612	405	217	82	11	6	73	269	467	674		
	CDD10.0	1140	0	0	0	5	64	178	315	343	197	37	2	0		
	CDD18.3	187	0	0	0	0	1	9	67	91	19	0	0	0		
	CDH23.3	777	0	0	0	0	16	53	238	396	74	0	0	0		
	CDH26.7	118	0	0	0	0	3	6	33	68	8	0	0	0		
(14) Wind	WSAvg	3.1	2.9	3.1	3.3	3.3	3.5	3.3	3.1	2.7	2.6	2.8	3.0	3.0		
(15) Precipitation	PrecAvg	1485	140	110	88	85	88	101	155	147	158	114	141	157		
	PrecMax	1741	214	187	145	136	131	160	269	284	302	173	253	288		
	PrecMin	1183	81	42	37	35	48	61	96	63	69	45	80	80		
	PrecStd	151	32	34	30	29	25	23	45	55	64	33	43	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	5.9	9.8	19.6	25.3	27.2	29.6	30.6	27.7	20.5	15.6	8.4		
		MCWB	2.2	3.4	5.3	11.6	16.0	19.1	23.3	24.3	22.0	15.5	11.7	6.4		
	2%	DB	2.0	3.6	7.6	16.1	22.0	24.8	27.4	28.5	25.3	18.7	13.2	5.4		
		MCWB	0.6	1.5	4.4	10.1	14.6	17.9	22.2	23.1	20.4	14.7	10.3	3.8		
	5%	DB	0.7	2.0	5.9	13.2	19.7	23.0	25.8	26.9	23.7	17.3	11.4	3.6		
		MCWB	-0.6	0.3	3.3	8.3	12.9	17.2	21.5	22.3	19.5	13.8	8.9	2.1		
	10%	DB	-0.6	0.7	4.4	10.9	17.6	21.3	24.4	25.5	22.3	15.9	9.7	2.0		
		MCWB	-1.8	-0.8	2.0	7.0	12.1	16.7	20.7	21.5	18.7	12.7	7.6	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	4.0	6.3	12.9	16.7	20.5	24.2	25.1	23.1	17.0	12.6	6.5		
		MCDB	3.4	5.6	8.5	17.8	22.7	24.7	27.9	29.4	26.7	18.6	14.8	8.0		
	2%	WB	0.9	1.8	4.9	10.6	15.2	19.2	23.1	23.9	21.8	15.6	10.7	4.0		
		MCDB	1.8	3.3	7.1	15.1	20.5	22.9	26.0	27.2	24.2	17.7	12.7	5.3		
	5%	WB	-0.4	0.6	3.6	8.9	14.0	18.2	22.3	23.1	20.6	14.4	9.2	2.2		
		MCDB	0.6	1.9	5.6	12.6	18.2	21.8	24.9	25.8	22.7	16.7	11.1	3.4		
	10%	WB	-1.8	-0.8	2.2	7.3	12.9	17.2	21.5	22.3	19.2	13.1	7.6	0.7		
		MCDB	-0.7	0.6	4.1	10.5	16.6	20.5	23.7	24.8	21.7	15.5	9.6	1.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.7	7.9	8.9	9.9	8.8	7.5	7.7	9.6	9.7	7.3	6.2		
		MCDBR	8.3	9.5	9.9	12.9	14.4	12.0	9.5	9.4	10.5	11.5	10.8	8.5		
	5% WB	MCWBR	7.4	8.0	7.3	7.7	7.7	6.2	4.9	4.9	6.0	7.8	8.3	7.4		
		MCWBR	7.9	9.2	9.2	11.7	12.1	9.8	7.8	7.8	8.4	9.9	10.3	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.342	0.392	0.459	0.519	0.535	0.514	0.498	0.465	0.410	0.402	0.386	0.361		
	taud		2.019	1.879	1.791	1.787	1.846	1.980	2.097	2.211	2.314	2.191	2.128	2.022		
	Ebn at Noon		746	765	762	753	755	774	781	796	816	761	696	678		
	Edh at Noon		113	155	192	208	202	177	156	135	113	113	102	104		
(44) All-Sky Solar Radiation	RadAvg		1.21	1.93	3.23	4.48	5.05	4.88	4.37	4.27	3.71	2.65	1.48	1.01		
	RadStd		0.09	0.18	0.27	0.46	0.47	0.47	0.46	0.34	0.37	0.16	0.15	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page